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Lecturer

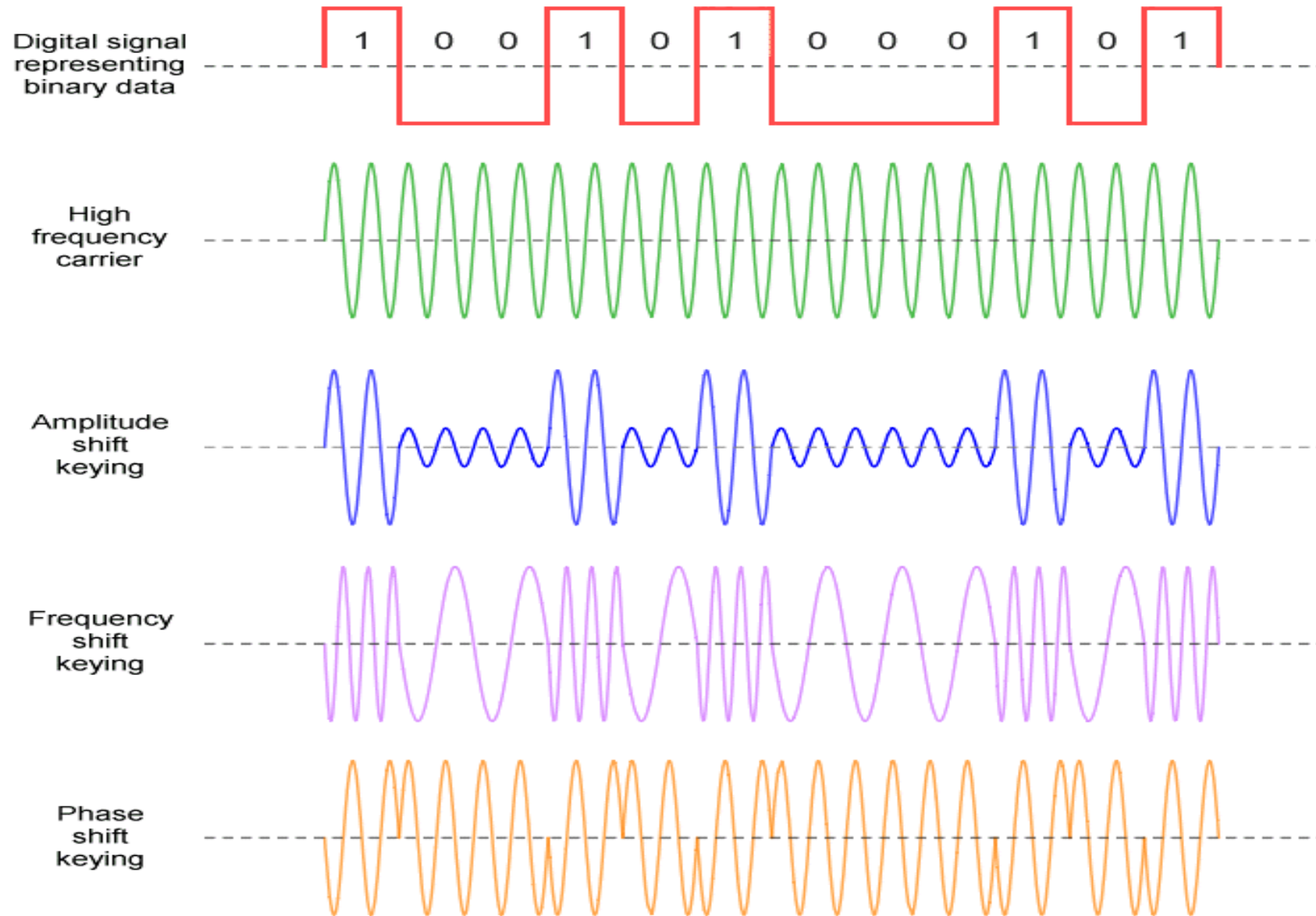
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ASK, FSK & PSK



AMPLITUDE SHIFT KEYING

Advantages:

- Very simple implementation
- Low cost and low power consumption
- Band efficient

Disadvantages:

- Highly susceptible to noise
- Not suitable for long distance or wireless communication

Applications:

- Optical fiber communication
- Short range data communication like sensor data
- IR remote control

PHASE SHIFT KEYING

Advantages:

- Better power efficiency than ASK & FSK
- Excellent noise immunity
- More bandwidth efficient

Disadvantages:

- Requires phase synchronization
- Sensitive to phase errors
- More complex transmitter and receiver

Applications:

- Wireless communication such as wifi, cellular system
- Satellite communication
- Digital broadcasting

FREQUENCY SHIFT KEYING

Advantages:

- Constant amplitude hence more efficient than ASK
- Less error prone
- Doesn't require phase synchronization

Disadvantages:

- Requires larger bandwidth
- Low data rate
- More complex than ASK

Applications:

- Telephone lines
- Low frequency systems
- Bluetooth

THINK

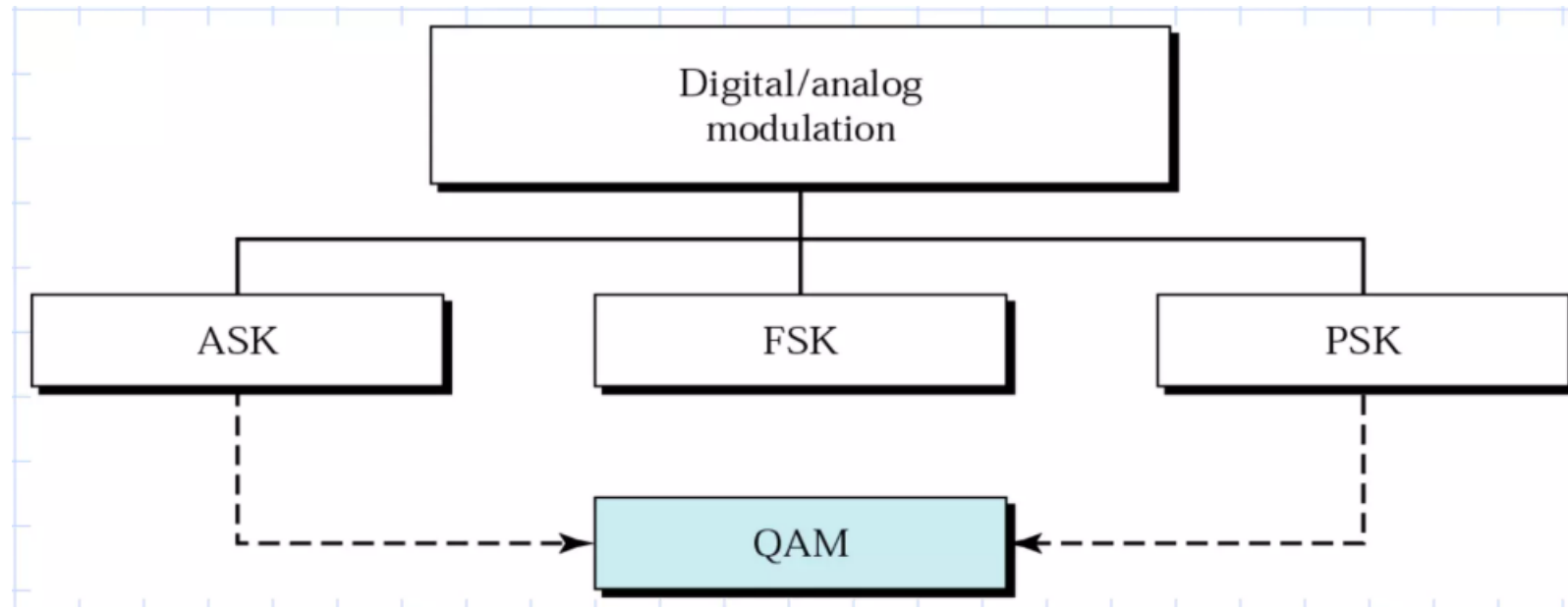
First Scenario: You are designing a very low-cost, simple transmitter for a short-range system. Which modulation technique would you choose?

ASK is simplest and cheapest to implement and easy to detect.

Second Scenario: A short-range radio link must send low-speed digital data in a moderately noisy environment. The system should be reliable but also cost effective. Justify which one would you choose ASK/FSK/PSK?

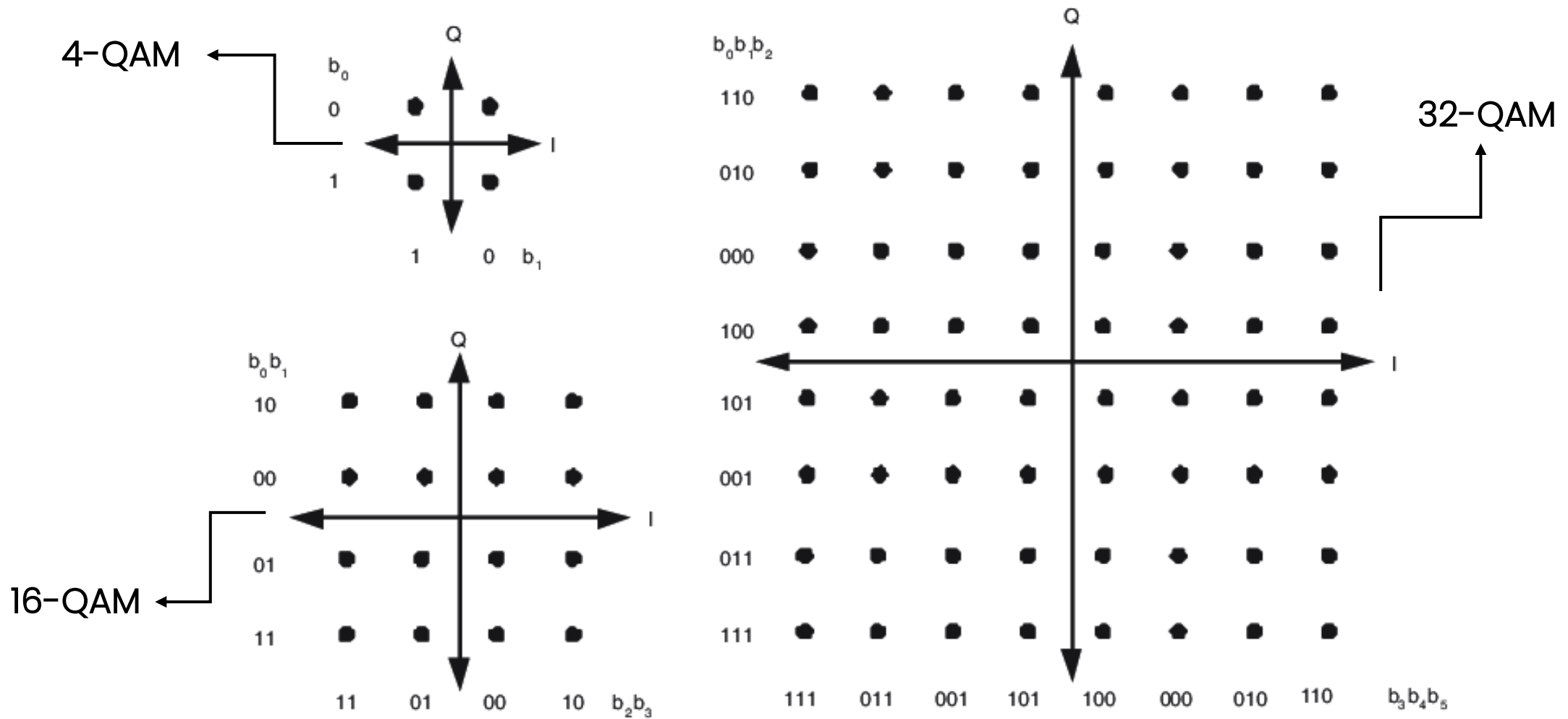
FSK has better noise immunity than ASK and is simpler than PSK.

QUADRATURE AMPLITUDE MODULATION

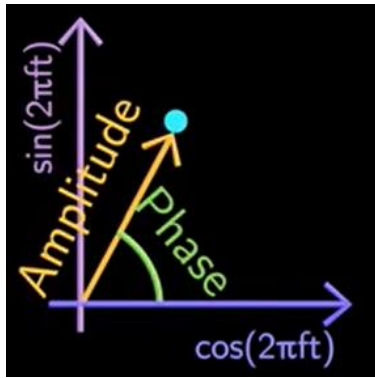


QAM is a modulation technique that combines ASK and PSK where both phase and amplitude of carrier changes while conveying the data signal.

QUADRATURE AMPLITUDE MODULATION

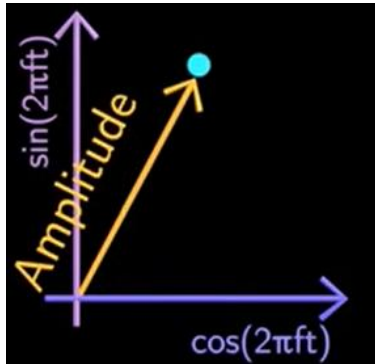
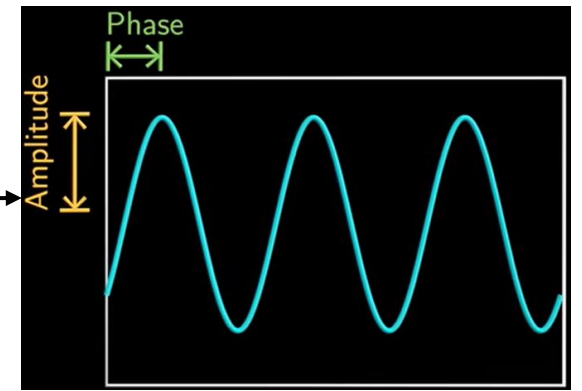


Constellation Diagram

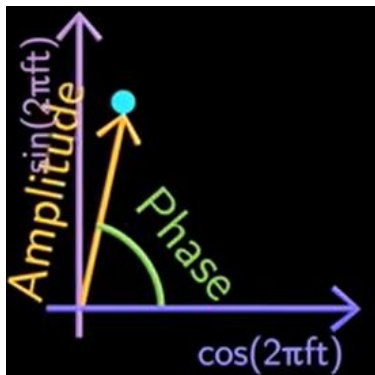
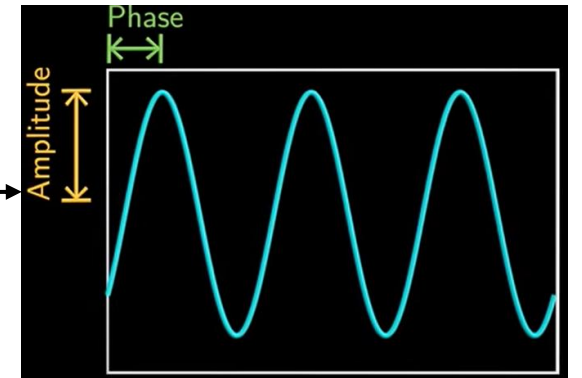


A data point can be represented on constellation diagram.

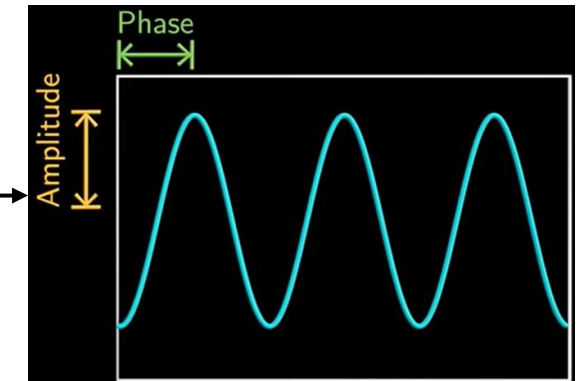
Amplitude= distance from origin
Phase= angle with x-axis



Increased distance represent higher amplitude and decreasing it will represent smaller signal amplitude

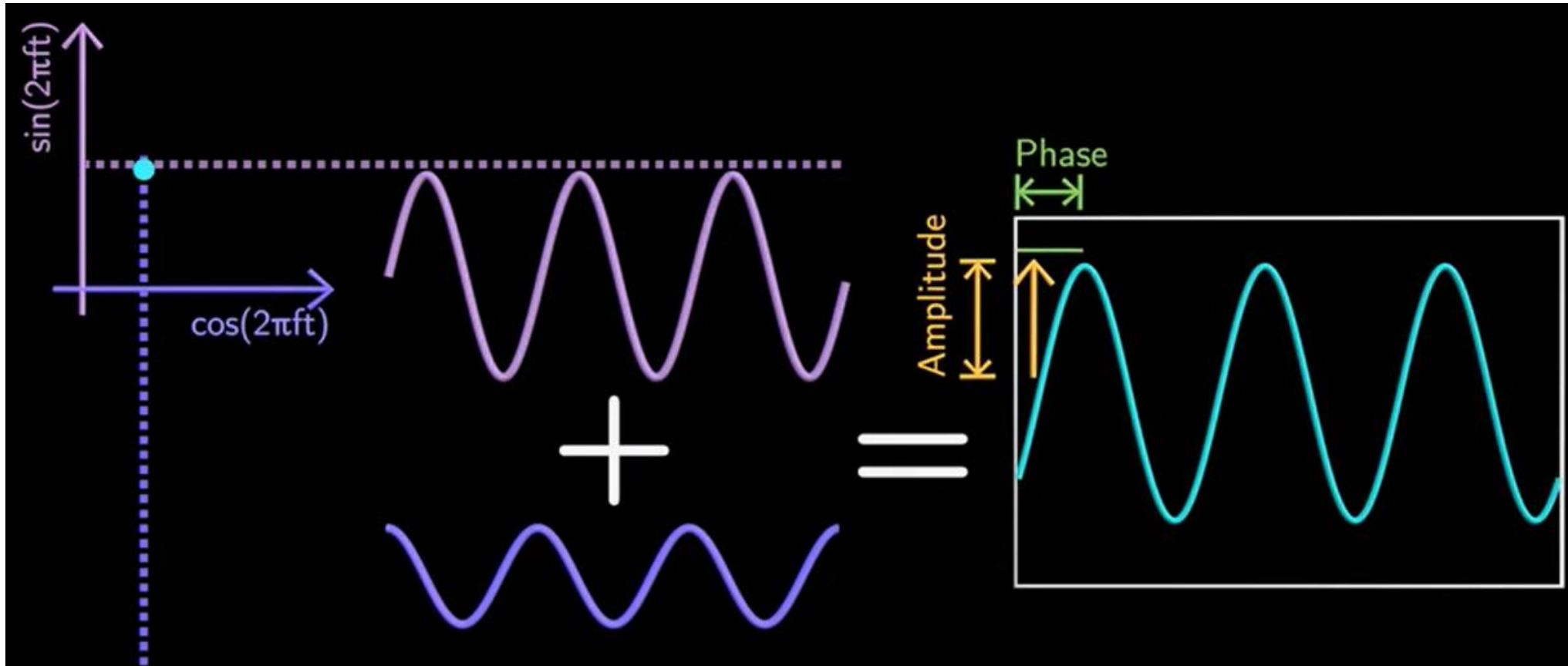


Phase changes with changing angle with x-axis



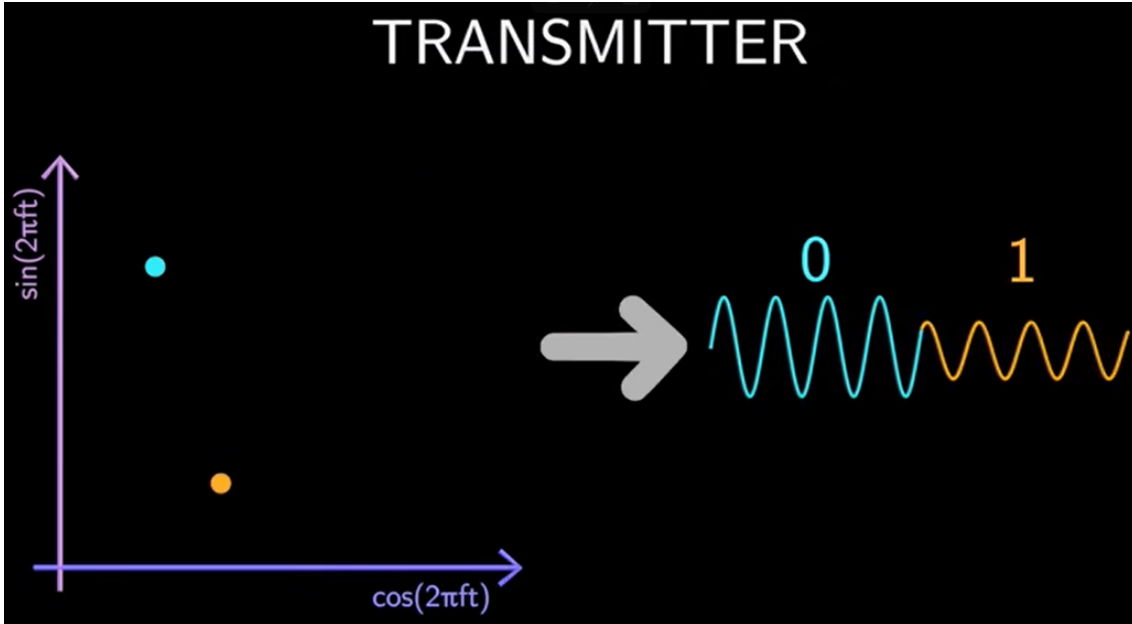
QUADRATURE AMPLITUDE MODULATION

The two components of the data point represent the digital signal to analog signal.



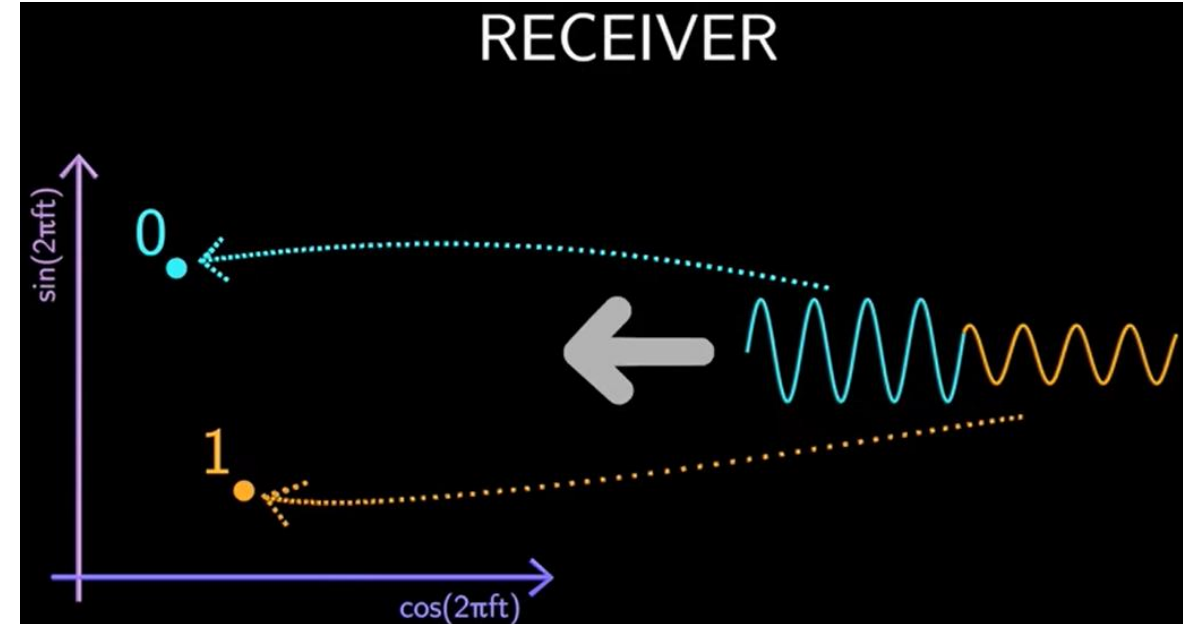
QUADRATURE AMPLITUDE MODULATION

TRANSMITTER



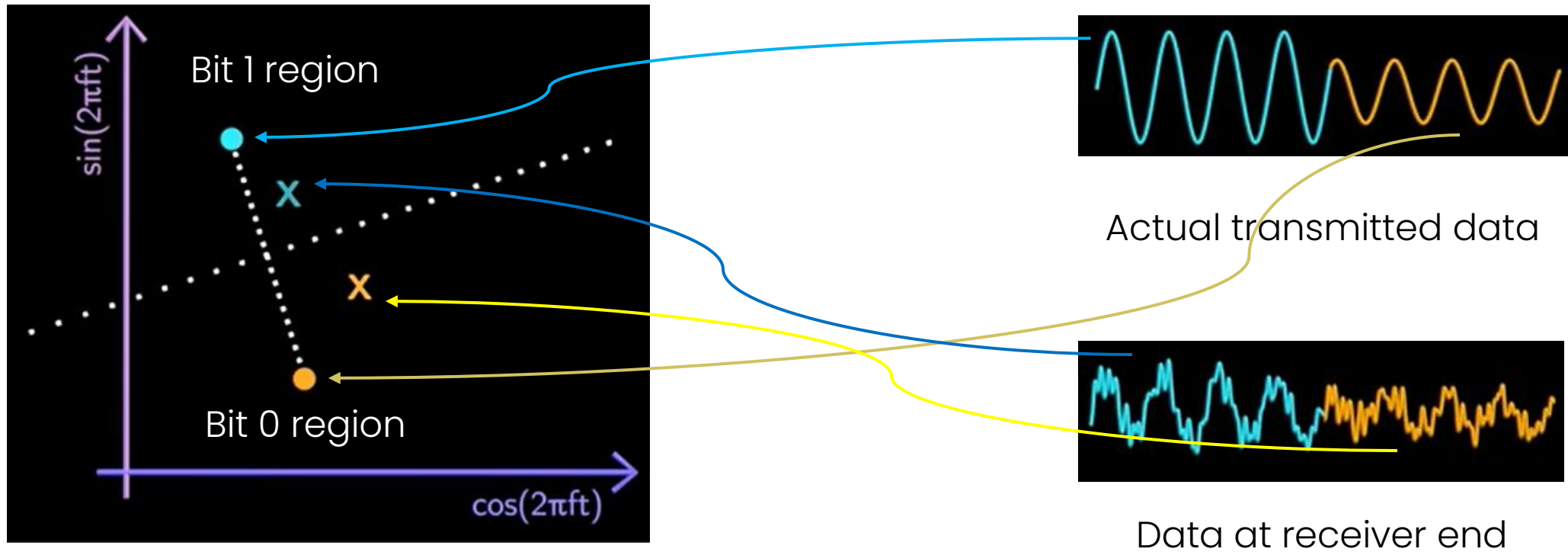
Transmitter takes the horizontal and vertical components of the data point to convert it to analog signal for transmission.

RECEIVER



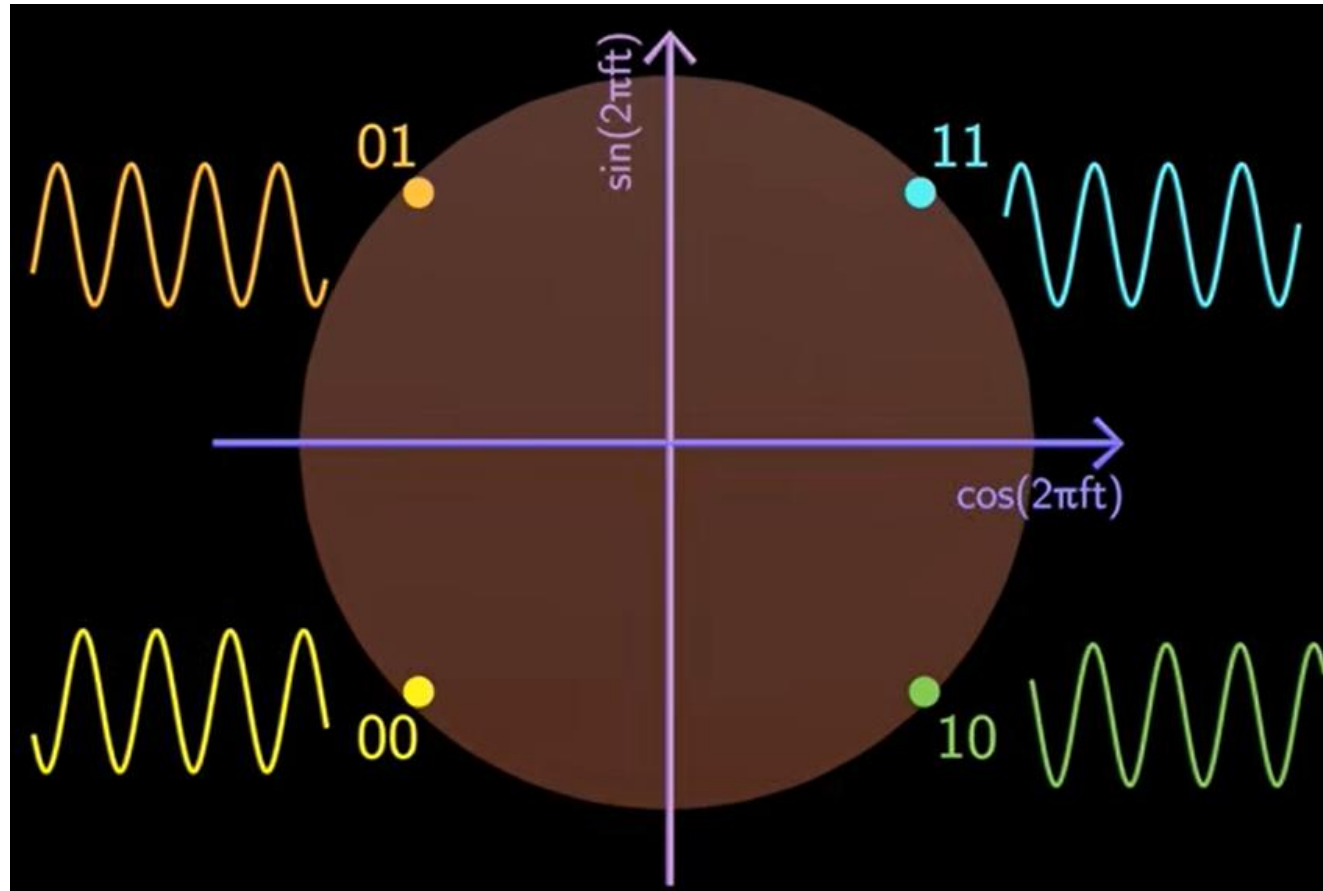
At receiver end it multiplies the signal with sine wave and integrates to get vertical component and cosine wave to retrieve horizontal component.

QUADRATURE AMPLITUDE MODULATION



QUADRATURE AMPLITUDE MODULATION

- Fitting constellation points far apart reduces chances of error.
- Stable and predictable system enables to pack more bits in a signal.
- More bits representing a signal increases spectral efficiency
- Easily decoded at receiver end.



QUADRATURE AMPLITUDE MODULATION

With increase number of bit QAM gets higher data rate and better spectral efficiency but noise tolerance decreases with increasing complexity of the receiver.

Advantages:

- Flexible and scalable
- Efficient use of bandwidth
- High efficiency and data rate

Disadvantages:

- Sensitive to noise
- Complex transmitter and receiver
- Requires power amplifiers

Applications:

- WiFi
- 4G, 5G
- Digital TV broadcasting



THANK YOU

